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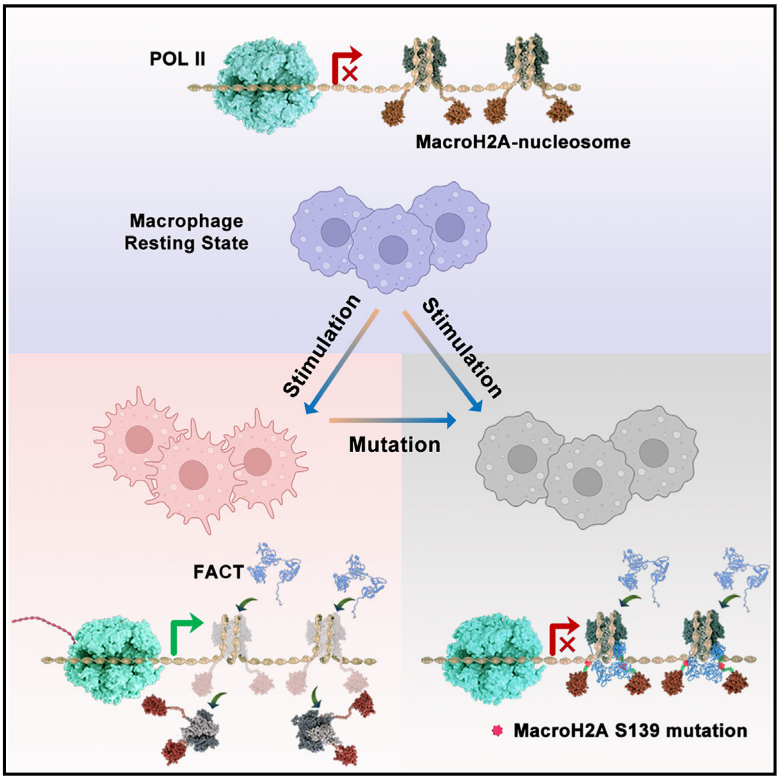
FACT mediates the depletion of macroH2A1.2 to expedite gene transcription

Graphical abstract



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In brief

Ji et al. demonstrate that although human macroH2A1.2 does not affect nucleosome stability or folding dynamics, it blocks FACT's function in nucleosome maintenance via key residue S139 and regulates the transcriptional response of macrophages. This study provides insights into the interplay between macroH2A1.2 and FACT at the nucleosome level.

Highlights

- Neither the macro nor linker domain of macroH2A1.2 affects the nucleosome stability
- MacroH2A1.2 blocks FACT's maintenance function at nucleosome level
- MacroH2AS139 is the key residue to switch the FACT-mediated depletion of macroH2A1.2
- FACT-mediated depletion of macroH2A1.2 modulates the immune response of macrophages

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FACT mediates the depletion of macroH2A1.2 to expedite gene transcription

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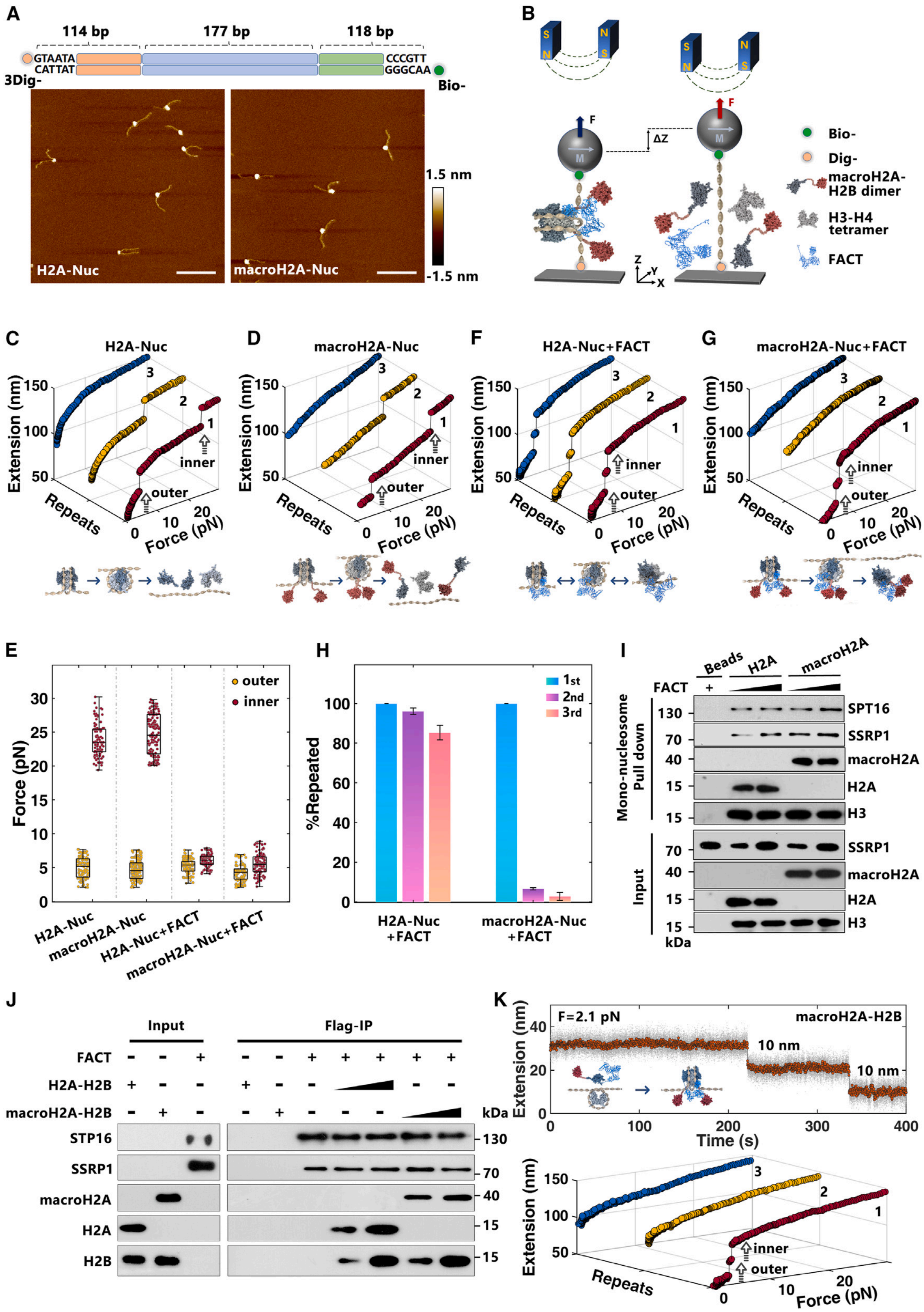
SUMMARY

The histone variant macroH2A is generally linked to transcriptionally inactive chromatin, but how macroH2A regulates chromatin structure and functions in the transcriptional process remains elusive. This study reveals that while the integration of human macroH2A1.2 into nucleosomes does not affect their stability or folding dynamics, it notably hinders the maintenance of facilitates chromatin transcription's (FACT's) function. We show that FACT effectively diminishes the stability of macroH2A1.2-nucleosomes and expedites their depletion subsequent to the initial unfolding process. Furthermore, we identify the residue S139 in macroH2A1.2 as a critical switch to modulate FACT's function in nucleosome maintenance. Genome-wide analyses demonstrate that FACT-mediated depletion of macroH2A-nucleosomes allows the correct localization of macroH2A, while the S139 mutation reshapes macroH2A distribution and influences stimulation-induced transcription and cellular response in macrophages. Our findings provide mechanistic insights into the intricate interplay between macroH2A and FACT at the nucleosome level and elucidate their collective role in transcriptional regulation and immune response of macrophages.

INTRODUCTION

The eukaryotic genome is packaged into chromatin by histones, which provides a structural platform for all genomic processes, including DNA replication, transcription, and damage repair. The accessibility of eukaryotic DNA is tightly regulated by the dynamic nature of chromatin states. Canonical histones (H2A, H2B, H3, and H4), typically encoded by multiple gene copies within chromosomal clusters, are expressed and incorporated into chromatin during the S phase, closely linked to DNA replication.[1–3] In addition to these canonical histones, certain specialized histones, known as histone variants, are encoded by individual genes. Their synthesis and chromatin incorporation occur independently of DNA replication.[2,3] These histone variants can replace their canonical counterparts in chromatin at specific genomic locations and represent a diverse epigenetic state for chromatin functions by directly altering the physical property and dynamics of the nucleosome, and/or by recruiting additional factors to modify the chromatin.

The histone variant macroH2A, the largest of the H2A variants, is highly conserved and widespread in vertebrates. It has a unique tripartite structure composed of an N-terminal histone-like region that is 64% identical to canonical H2A, an unstructured linker region, and a bulky C-terminal globular macro domain.[4] In mammalian cells, two isoforms, macroH2A1 and macroH2A2, are encoded by H2afy and H2afy2, respectively. H2afy further produces two splice variants, known as macroH2A1.1 and macroH2A1.2, with mutually exclusive exons (VIb and VIa, respectively) within the macro domain.[5] MacroH2A has been found to be enriched in heterochromatin, including the inactive X chromosome of female mammalian cells and senescence-associated heterochromatin foci.[6,7] Structural analysis showed that the L1 loop in the histone fold domain of macroH2A creates an inflexible L1-L1 interface between the two H2A-H2B dimers to stabilize the histone octamer in nucleosome state, with the non-histone region protruding out of the nucleosome core particle near the dyad DNA.[8,9] The linker region of macroH2A, with a similar amino acid composition to



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the C terminus of histone H1, has been shown to increase the protection of the entry/exit DNA site of nucleosomes from exonuclease III digestion and promote the condensation and fiber-fiber interaction of nucleosomal arrays.[10,11] *In vitro* investigations have found that macroH2A—primarily its non-histone region—hinders the access of transcription factors, p300-dependent histone acetylation, switch/sucrose non-fermentable (SWI/SNF) and ATP-utilizing chromatin assembly and remodeling factor (ACF) chromatin remodeling, and acts as an efficient repressor of RNA polymerase II (RNA Pol II) transcription.[12,13] In addition, macroH2A has been found to cooperate with other epigenetic factors, including the recruitment of histone deacetylase 1 and 2 (HDAC1/2), polycomb repressive complex 2 (PRC2), and heterochromatin protein 1 (HP1).[14–16] Furthermore, macroH2A has been revealed to undergo several post-translational modifications, including mono-ubiquitination at K115/116 and phosphorylation at T128/S137, which provide further refinement to regulate chromatin states and functions.[17–19] However, despite extensive efforts, the physiological roles of macroH2A variants remains elusive.

The histone chaperone facilitates chromatin transcription (FACT) complex, which is highly conserved among eukaryotes and essential for cell viability, was initially identified as a histone chaperone of H2A-H2B dimers.[20,21] FACT has been found to facilitate the progression of DNA and RNA polymerases on chromatin templates[22,23] as well as maintain the genome-wide integrity of chromatin.[24–30] Recently, we and others have revealed that FACT can tether both the $(H3\text{-}H4)_2$ tetramer and H2A-H2B dimer on DNA through its two subunits, suppressor of Ty 16 (SPT16) and structure-specific recognition protein-1 (SSRP1), which function distinctly but coordinately to mediate nucleosome assembly/disassembly, as demonstrated by *in vitro* structural and single-molecular investigations.[31,32] It is of great interest to explore how FACT's functions are regulated by different histone variants. Previous research has shown that FACT is required for the pruning of macroH2A from the transcribed chromatin region to establish and maintain the genomic localization of macroH2A.[33] Additionally, FACT has been found to promote the deposition of macroH2A1.2 at sites of replication-stress-induced DNA damage to form a protective chromatin environment.[34] However, the direct influence of macroH2A on FACT's functions remains to be understood.

In this work, we investigated the regulatory functions of macroH2A1.2 on nucleosome dynamics and their impact on the function of the FACT complex at the nucleosome level. We found that macroH2A, regardless of its macro or linker domain, does not affect the mechanical stability or folding kinetics of nucleosomes. Significantly, the incorporation of macroH2A into the nucleosome impairs FACT's ability to maintain nucleosome integrity. The binding of FACT can significantly weaken the stability of the macroH2A-nucleosome and quickly deplete the macroH2A-nucleosome following the first unfolding process. Furthermore, we identified the key residue S139 in the linker region of macroH2A, which plays a crucial role in FACT-mediated depletion of the macroH2A-nucleosome. The macroH2A$_{S139A}$ mutation can restore FACT's maintenance function at the nucleosome level. Genome-wide analyses indicated that FACT-mediated depletion of the macroH2A-nucleosome plays a significant role in regulating the cellular response of macrophages. The S139 mutation in macroH2A reshapes the position of macroH2A and regulates stimulation-induced transcription and the function of macrophages. Our study offers mechanistic insights into the intricate interplay between macroH2A and FACT at the nucleosome level and advances our understanding of their collective role in the immune response of macrophages.

RESULTS

MacroH2A1.2 does not affect the stability and folding kinetics of nucleosomes, but it blocks FACT's function in nucleosome maintenance

As a structurally unique histone variant that contains a big globular macro domain connected by an unstructured linker, how macroH2A affects the structure of the nucleosome remains to be elucidated. We directly investigate the effect of macroH2A1.2 on the mechanical stability and folding kinetics of the nucleosome by single-molecule magnetic tweezers. This approach, employing the single-molecule force spectroscopy, facilitates a precise analysis of structural dynamics through the direct manipulation of the nucleosome. The macroH2A-containing mono-nucleosomes were reconstituted on a 409-bp DNA containing one Widom 601 nucleosome position sequence in the middle and characterized by atomic force microscopy (AFM) imaging and gel shift analysis (Figures 1A, S1A, and S1B). The real-time trajectories of force-dependent conformational changes for each nucleosome were traced, with the stretching force ranged from 0.1 to 30 pN for one trajectory and repeated for at least three

Figure 1. MacroH2A1.2 does not affect the stability and folding kinetics of the nucleosome but blocks FACT's function in nucleosome maintenance

(A) AFM images of H2A-nucleosomes (left, H2A-Nuc) and macroH2A-nucleosomes (right, macroH2A-Nuc) reconstituted on the DNA template shown at the top. Biotin-labeled DNA (Bio-) and 3 digoxigenin-labeled DNA (3Dig-). Scale bars: 100 nm. Height: −1.5 to 1.5 nm.

(B) A schematic representation of single-molecule stretching experiment by magnetic tweezers (not to scale).

(C and D) The typical repeated stretching measurements of H2A-Nuc (C) and macroH2A-Nuc (D).

(E) The related statistical rupture forces for the outer and inner DNA wrap shown in (C) and (D) ($n = 93$ for C, $n = 150$ for D) and (F) and (G) ($n = 102$ for F, $n = 106$ for G).

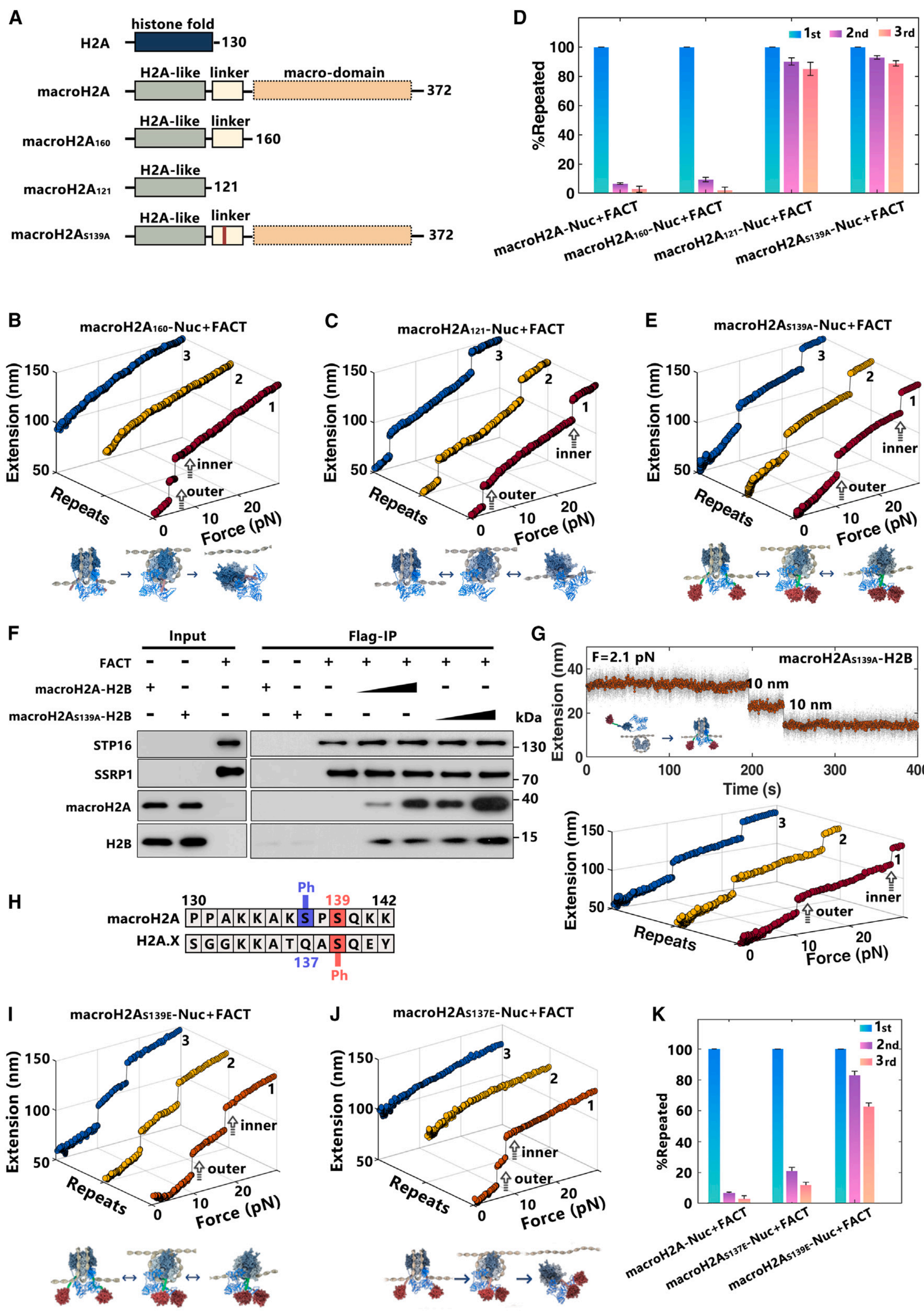
(F and G) The typical repeated stretching measurements of H2A-Nuc (F) and macroH2A-Nuc (G) with FACT.

(H) The statistical proportion of nucleosome maintenance in each of the three repeated stretching measurements for H2A- and macroH2A-nucleosomes with FACT ($n = 3$, see the analysis details in STAR Methods).

(I) Western blot analysis for the mono-nucleosome pull-down assay of H2A-Nuc (H2A) and macroH2A-Nuc (macroH2A), incubated with the purified FACT, $n = 3$.

(J) FLAG-IP analysis of the recombinant FACT with FLAG-SPT16 and SSRP1 incubated with recombinant H2A-H2B or macroH2A-H2B dimer, $n = 3$.

(K) The real-time sequential deposition of the two macroH2A-H2B dimers on the reconstituted tetrasome with the help of FACT (top), and the repeated stretching measurements of the same molecule after the deposition process (bottom).



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times for each nucleosome sample (Figure 1B). Typically, two structural transition steps at tensions of around 5 and 23 pN, corresponding to the unraveling of the outer and inner DNA wrap of the nucleosome, were observed in the first stretching cycle for both H2A- and macroH2A-nucleosomes (Figures 1C and 1D), consistent with previous results of ours[32,35,36] and others.[37,38] The rupture forces for the outer and inner wrap, as pivotal indicators of its mechanical stability, were statistically analyzed (Figure 1E), which revealed that macroH2A does not affect the mechanical stability of nucleosomes. In addition, in the following repeated stretching cycles, for both the H2A- and macroH2A-nucleosome, the typical two-step unfolding process of intact nucleosomes cannot be observed (Figures 1C, 1D, and S1C), which is due to the displacement of core histones from DNA when the nucleosome structure is totally disrupted.[39] The results indicated that macroH2A does not influence the mechanical stability and folding kinetics of the nucleosome.

Previous study has shown that FACT is required for the pruning of macroH2A from the transcribed chromatin region.[33] To determine how FACT directly functions on macroH2A-nucleosomes, we traced the structural transition of macroH2A mononucleosome incubated with the purified FACT complex (Figure S1D). In the presence of FACT, the canonical H2A-nucleosome was observed to be completely unfolded at much lower forces (<10 pN) and retain a similar force response in the repeated stretching cycles (Figures 1E and 1F), which indicated that FACT impairs the stability of the H2A-nucleosome and maintains its integrity, consistent with our previous results.[32] Intriguingly, for the macroH2A-nucleosome, FACT cannot maintain the integrity of nucleosome at all (Figure 1G). In the presence of FACT, the macroH2A-nucleosome is also disrupted at low forces (<10 pN), with two structural transition steps in the first stretching cycle (Figures 1E and 1G). However, totally different from the canonical H2A-nucleosome, almost no structural transition can be observed in the following repeated stretching cycles of the macroH2A-nucleosome, which implies that the nucleosome is totally depleted after the first stretching cycle. The properties for nucleosome maintenance were statistically analyzed (Figure 1H) and suggested that macroH2A greatly blocks FACT's function in nucleosome maintenance. In addition, the effect of macroH2A on the binding capacity of FACT on the nucleosome was investigated (Figure 1I). The mono-nucleosome pull-down assays showed that the incorporation of macroH2A does not impair, but slightly enhances, the binding of the purified FACT complex onto the nucleosome (Figure 1I). Altogether, the results demonstrated that macroH2A promotes the binding of the FACT complex onto the nucleosome, which greatly impairs macroH2A-nucleosome stability and quickly depletes the nucleosome at low force.

We then investigated how macroH2A regulates the chaperone property of FACT. The recombinant H2A-H2B dimer or macroH2A-H2B dimer was incubated with the purified recombinant FACT complex, composed of FLAG-SPT16 and SSRP1. The binding properties of FACT with the H2A-H2B dimer or macroH2A-H2B dimer were examined by FLAG-immunoprecipitation (FLAG-IP) assays, which revealed that FACT binds both the H2A-H2B dimer and macroH2A-H2B dimer with similar efficiency (Figure 1J). The results indicated that macroH2A does not affect the binding properties of FACT with the H2A-H2B dimer. The effect of macroH2A on the properties of FACT-mediated nucleosome assembly were investigated by magnetic tweezers. Repeated force-extension measurements of the reconstituted mono-$(H3\text{-}H4)_2$ tetrasome revealed the typical, irreversible one-step transition at a force of around 16 pN, corresponding to the disruption of the tetrasome (Figure S1E, top). For the tetrasome incubated with FACT, a reversible one-step transition at a lower force of about 8 pN was observed (Figure S1E, bottom), indicating that FACT impairs the stability of the tetrasome and maintains its integrity. We then traced the *in situ* deposition of the macroH2A-H2B dimer onto the tetrasome. We found after the addition of FACT and macroH2A-H2B dimers into the anchored tetrasome, two sequential 10-nm deposition steps were clearly observed, which corresponds to the sequential deposition of the two dimers (Figure 1K, top). After the real-time deposition process, we carried out the repeated force-extension measurements for the same molecule, and a typical irreversible two-step unfolding pathway of intact macroH2A-nucleosomes was observed (Figure 1K, bottom). The results revealed that FACT can deposit two macroH2A-H2B dimers onto the tetrasome to form an intact macroH2A-nucleosome, as happens with the canonical H2A-H2B dimers (Figure S1F).

MacroH2A1.2 blocks FACT's function in nucleosome maintenance by the key residue S139

We then ascertained the specific domain of macroH2A1.2 that blocks FACT's function in nucleosome maintenance. As compared with the canonical H2A, macroH2A contains a unique tripartite structure composed of an N-terminal histone-like region, an unstructured linker region, and a bulky C-terminal macro domain[4] (Figures 2A and S2A). The linker region of macroH2A,

Figure 2. MacroH2A1.2 blocks FACT's function in nucleosome maintenance by the key residue S139

(A) Schematic representation of the H2A, truncated and mutated macroH2A1.2 proteins.

(B and C) The typical repeated stretching measurements of $macroH2A_{160}$-Nuc (B) and $macroH2A_{121}$-Nuc (C) with FACT.

(D) The statistical proportion of nucleosome maintenance in each of the three repeated stretching measurements for macroH2A-Nuc, $macroH2A_{160}$-Nuc, $macroH2A_{121}$-Nuc, and $macroH2A_{S139A}$-Nuc with FACT.

(E) The typical repeated stretching measurements of $macroH2A_{S139A}$-Nuc with FACT.

(F) FLAG-IP analysis of the FLAG-SPT16 and SSRP1 incubated with macroH2A-H2B or $macroH2A_{S139A}$-H2B dimer, $n = 3$.

(G) The real-time sequential deposition of the two $macroH2A_{S139A}$-H2B dimers on the tetrasome with the help of FACT (top), and the repeated stretching measurements of the same molecule after the deposition process (bottom).

(H) Amino acid sequence comparison between the human macroH2A1 and H2A.X (residues 130–142 amino acid [aa]), with key histone modifications labeled.

(I and J) The typical repeated stretching measurements of the $macroH2A_{S139E}$-Nuc (I), $macroH2A_{S137E}$-Nuc (J) with FACT.

(K) The statistical proportion of nucleosome maintenance in each of the three repeated stretching measurements for macroH2A-Nuc, $macroH2A_{S137E}$-Nuc, $macroH2A_{S139E}$-Nuc with FACT.

with a similar amino acid composition to the C terminus of histone H1, has been shown to increase protection of the entry/exit DNA site of the nucleosome from exonuclease III digestion.[10] We first determined the direct effect of the linker region or macro domain of macroH2A on the stability and folding kinetics of the nucleosome. The truncated macroH2A proteins, including macroH2A$_{160}$, with macro domain truncated, and macroH2A$_{121}$, with both the macro and linker domains truncated, were purified and reconstituted into the nucleosome (Figure S2B). The real-time trajectories of force-dependent conformational changes for each macroH2A$_{160}$- and macroH2A$_{121}$-nucleosome were traced (Figures S2C and S2D). The results showed that, as compared with the canonical H2A-nucleosome, neither the macro domain nor the linker domain of macroH2A affect the stability and folding kinetics of the nucleosome (Figure S2E). We then investigated whether the macro and linker domains of macroH2A influence the functioning of FACT. The structural transition of macroH2A$_{160}$ and macroH2A$_{121}$ mono-nucleosome, incubated with the purified FACT complex, were traced by magnetic tweezers (Figures 2B and 2C), with the properties for nucleosome maintenance statistically analyzed (Figure 2D). We found that macroH2A$_{160}$, with the macro domain deleted, does not affect FACT's depletion function on the macroH2A-nucleosome. However, upon further deletion of the linker domain, macroH2A$_{121}$ restored FACT's function in nucleosome maintenance. The results indicated that the linker domain of macroH2A plays an essential role in blocking FACT's function in nucleosome maintenance. Deeper truncation investigations in the linker domain revealed that the residue range from 137 to 142 in macroH2A plays an essential role in blocking the maintenance function of FACT (Figures S2F–S2I). Further point-mutation analysis in the residues ranging from 137 to 142 of macroH2A identify the key residue S139 of macroH2A (Figures S2J–S2L). Single point mutations at the S139 of macroH2A (macroH2A$_{S139A}$) were purified and reconstituted into the nucleosome (Figure S2M). The macroH2A$_{S139A}$ does not affect the stability and folding kinetics of the macroH2A-nucleosome (Figure S2N). But, interestingly, in the presence of the FACT complex, the macroH2A$_{S139A}$-nucleosome becomes much more robust, with a much higher reversibility as compared with those of the wild-type macroH2A-nucleosome (Figures 2D and 2E). The results revealed that the key residue S139 in macroH2A plays an essential role in regulating FACT's function in nucleosome maintenance. In addition, the effect of macroH2A$_{S139A}$ mutation on the nucleosome binding capacity of FACT was examined by mono-nucleosome pull-down assay (Figures S2O and S2P), which showed that the S139A mutation has a minor effect on the binding of the purified FACT complex to the macroH2A-nucleosome.

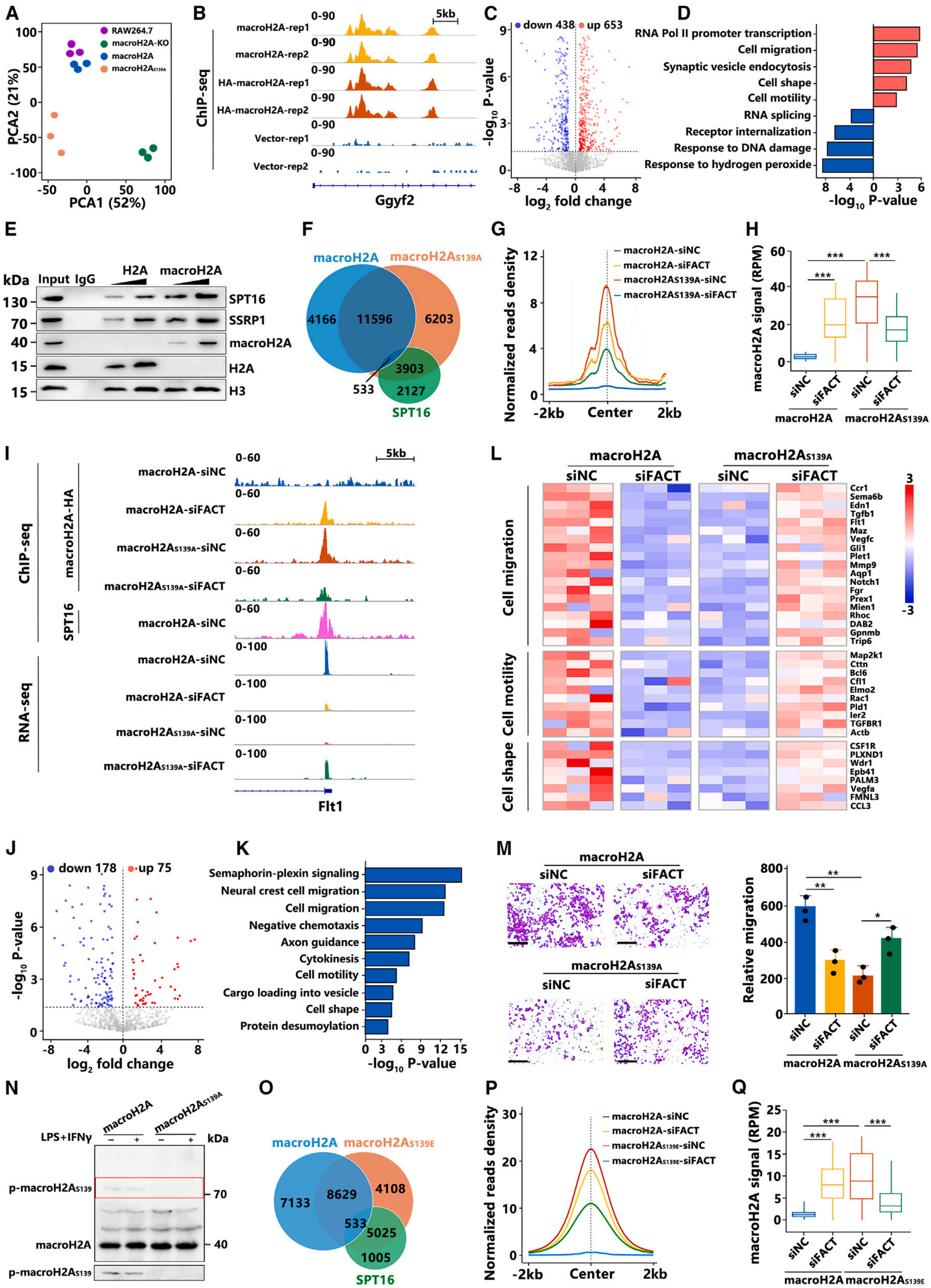
We further assessed how the S139A mutation in macroH2A influences the H2A-H2B chaperone properties of the FACT complex. The FLAG-IP assay showed that the S139A mutation enhances the binding of FACT on macroH2A-H2B (Figure 2F). The *in situ* deposition assay revealed that, after the addition of FACT and macroH2A$_{S139A}$-H2B dimers into the anchored tetrasome, two sequential 10-nm deposition steps were clearly observed, which corresponds to the sequential deposition of the two dimers (Figure 2G, top). In addition, after the real-time deposition process, we carried out the repeated force-extension measurement for the same molecule, and the typical reversible two-step unfolding pathway of the intact macroH2A$_{S139A}$-nucleosome was observed (Figure 2G, bottom). The results denote that the S139A mutation has a subtle effect on the H2A-H2B chaperone property of FACT.

The key residue S139 is highly conserved in the other H2A variant of H2A.X (the sequence alignment in Figure 2H), which can be chemically modified *in vivo*, γH2A.X (H2AX phosphorylation at S139). In fact, the phosphorylation of macroH2A at S139 has also been observed by mass spectrometry in UV-damaged M059K glioblastoma cells,[40] but its function remains unclear. The effect of macroH2A bearing the phosphor-mimicking glutamic acid mutation at S139 (S139E) on the functioning of FACT was examined, which revealed that the S139E in macroH2A can reverse the depletion function of FACT on macroH2A-nucleosomes (Figures 2I and S2Q). However, as a control, S137E, the phosphor-mimicking mutation at residue S137 whose phosphorylation has been revealed by mass spectrometry,[19] does not affect the function of FACT on macroH2A-nucleosomes (Figures 2J, 2K, and S2R). The results suggested that the depletion or maintenance of macroH2A by the FACT complex may be precisely regulated by dynamic modification at the key residue S139 *in vivo*.

S139 is the key residue to switch FACT-mediated depletion of macroH2A-nucleosomes and reshape the genomic localization of macroH2A in macrophages

Our *in vitro* investigation revealed that the binding of FACT on macroH2A-nucleosomes will significantly weaken the stability of the nucleosome and quickly deplete the macroH2A-nucleosome following the first unfolding process. Furthermore, we identified the critical residue S139 in macroH2A as the key site to switch FACT's function in nucleosome maintenance/depletion. Previous studies have revealed that macroH2A functions to mediate the gene expression in response to environmental stimuli, including pro-inflammatory signals or a tumor-associated microenvironment.[41] We examined the function of macroH2A and its dynamic depletion by FACT in RAW264.7 cells (a murine leukemic monocyte/macrophage cell line). We generated the endogenous macroH2A knockout (KO) RAW264.7 cell line through CRISPR targeting of H2afy (macroH2A-KO cells), and "rescued" the macroH2A-KO cells with hemagglutinin (HA)-tagged wild-type macroH2A (macroH2A cells) or HA-tagged macroH2A$_{S139A}$ (macroH2A$_{S139A}$ cells) (Figure S3A). Principal component analysis on the RNA sequencing (RNA-seq) data revealed that the wild-type RAW264.7 cells and macroH2A cells cluster together, whereas macroH2A-KO cells cluster independently with high PC1, consistent with a common rescue phenotype (Figure 3A). The rescue of the specific genes affected in the macroH2A-KO cells by the wild-type macroH2A re-expression were shown in Figure S3B.

Comparing the macroH2A-KO cells with macroH2A cells, RNA-seq analysis showed that statistically significant representations of genes involved in "transcription," "cell migration," "cell motility," and "cell shape" were regulated by macroH2A, as revealed by Gene Ontology (GO) analysis (Figures S3C and S3D). To further identify the genes directly regulated by macroH2A,



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we performed chromatin immunoprecipitation followed by sequencing (ChIP-seq) analysis of macroH2A using anti-HA antibodies (Figures 3B and S3E) to define the macroH2A-bound genes overlapped with macroH2A ChIP-seq signals in regions including the transcription start site (TSS), promoter, intron, and exon. The ChIP-seq data agree well with those conducted using the anti-macroH2A antibodies (Figures 3B and S3F), targeting the different epitopes at the C-terminal macro domain of macroH2A, which excludes the possibility of epitope effects. RNA-seq analysis on the transcriptomic profile of the macroH2A-KO and macroH2A cells revealed 653 macroH2A-bound genes that are upregulated by macroH2A-KO (Figure 3C). GO terms revealed that these genes are functionally associated with cell migration, cell shape, and cell motility (Figure 3D). All the results indicated that the histone variant macroH2A plays an important role in regulating the immune functions of macrophages.

To examine the function of FACT on the macroH2A-nucleosome, we performed mono-nucleosome pull-down assays, which revealed that, as compared with the canonical H2A-nucleosome, FACT prefers to bind the macroH2A-nucleosome (Figure 3E), consistent with our analysis in the purified *in vitro* system (Figure 1I). ChIP-seq analysis of SPT16, a subunit of the FACT complex, was conducted in macroH2A cells (Figure S3G). By comparing the genome-wide binding of macroH2A versus FACT in macroH2A cells, we found that only ~3% (533 regions) of the macroH2A-bound regions are colocalized with FACT (Figure 3F), which agrees with our *in vitro* analysis that FACT functions to deplete macroH2A. Interestingly, for the mutant macroH2A$_{S139A}$, although it does not affect the binding capacity of FACT to the macroH2A-nucleosome (Figure S3H), ChIP-seq analysis revealed that more 3,903 macroH2A$_{S139A}$-bound regions are colocalized with FACT (Figure 3F). The results indicated that in these 3,903 regions, the enriched FACT complex can deplete the wild-type macroH2A but cannot deplete the mutant macroH2A$_{S139A}$. To further test this hypothesis, we knocked down the endogenous FACT expression (Figures S3I and S3J). The knockdown of FACT does not affect the total levels of wild-type macroH2A and macroH2A$_{S139A}$ (Figure S3K) but causes slightly more macroH2A-bound at the chromatin fraction (Figure S3L) and greatly affects the localizations of macroH2A at these 3,903 regions. The ChIP-seq analysis revealed that, in macroH2A cells, macroH2A is depleted by FACT at these regions (blue line in Figures 3G and 3H); knockdown of FACT in the cells causes the massive enrichment of macroH2A at these sites (yellow line in Figures 3G and 3H). However, in the macroH2A$_{S139A}$ cells, the enriched FACT can maintain macroH2A$_{S139A}$ at these regions (orange line in Figures 3G and 3H); knockdown of FACT in the cells causes the massive loss of macroH2A$_{S139A}$ at these sites (green line in Figures 3G and 3H). The typical tracks for the RNA-seq and ChIP-seq of FACT, macroH2A and macroH2A$_{S139A}$ in the related cells, were shown in Figure 3I. The results agree well with our *in vitro* analysis that FACT functions to deplete macroH2A while the S139A mutation in macroH2A reverses this function. By coupling the 3,903 regions with the RNA-seq analysis on the transcriptomic profile of macroH2A and macroH2A$_{S139A}$ cells, we defined 178 downregulated genes bound both with macroH2A$_{S139A}$ and FACT (Figure 3J). GO analysis revealed that these genes are functionally associated with the regulation of cell migration, cell motility, and cell shape (Figures 3K and 3L). To further investigate the biological functions for FACT-mediated depletion of macroH2A, we measured cell migration ability. We found that, compared with macroH2A cells, migration abilities were significantly reduced in macroH2A cells with FACT knockdown and in macroH2A$_{S139A}$ cells, whereas the migration abilities were rescued in macroH2A$_{S139A}$ cells with FACT knockdown (Figure 3M). All the results indicated that the binding of FACT on macroH2A-nucleosomes destabilizes and depletes the nucleosome, but macroH2A$_{S139A}$ reverses FACT-dependent depletion of macroH2A-nucleosomes and impairs the function of macrophages.

But for the 11,596 regions in Figure 3F, where the non-FACT-bound region overlapped with both wild-type macroH2A and

Figure 3. MacroH2A$_{S139A}$ reverses FACT-dependent depletion of macroH2A-nucleosomes in macrophages

(A) Transcriptomic principal-component analysis (PCA) of the RAW264.7, macroH2A-KO, macroH2A, and macroH2A$_{S139A}$ cells, $n = 3$.

(B) The ChIP-seq tracks for macroH2A at specific genomic regions in macroH2A and vector cells.

(C) Volcano plot of transcriptomic changes for macroH2A-bound genes identified by macroH2A ChIP-seq analysis between the macroH2A-KO and macroH2A cells. Significant genes (fold-change > 2, $p < 0.05$). Blue: downregulated genes; red: upregulated genes.

(D) Gene Ontology (GO) analysis in biological pathway category for the upregulated (red) and downregulated (blue) genes defined in (C) by DAVID (ranged by p value).

(E) Western blot analysis for the mono-nucleosome IP assay of H2A and HA-macroH2A in macroH2A cells.

(F) Venn diagram analysis on the ChIP-seq data for macroH2A and SPT16 in macroH2A cells, macroH2A$_{S139A}$ in macroH2A$_{S139A}$ cells.

(G and H) Density plot (G) and boxplots (H) analysis on the read density of macroH2A or macroH2A$_{S139A}$ at 3,093 peak regions in their corresponding cells.

(I) The ChIP-seq tracks (top) and the RNA-seq tracks (bottom) at specific genomic regions, which are bound with macroH2A or SPT16 in their corresponding cells.

(J) Volcano plot of transcriptomic changes for 3,903 peak-region-regulated genes between macroH2A$_{S139A}$ and macroH2A cells.

(K) GO analysis for the downregulated genes defined in (J) by DAVID.

(L) Heatmap of murine genes associated with GO terms defined in (K). Expression values are normalized row-wise as Z scores.

(M) Cell migration analysis by transwell invasion assays (left) and statistical analysis (right) in their corresponding cells, $n = 3$. Scale bars: 500 μm.

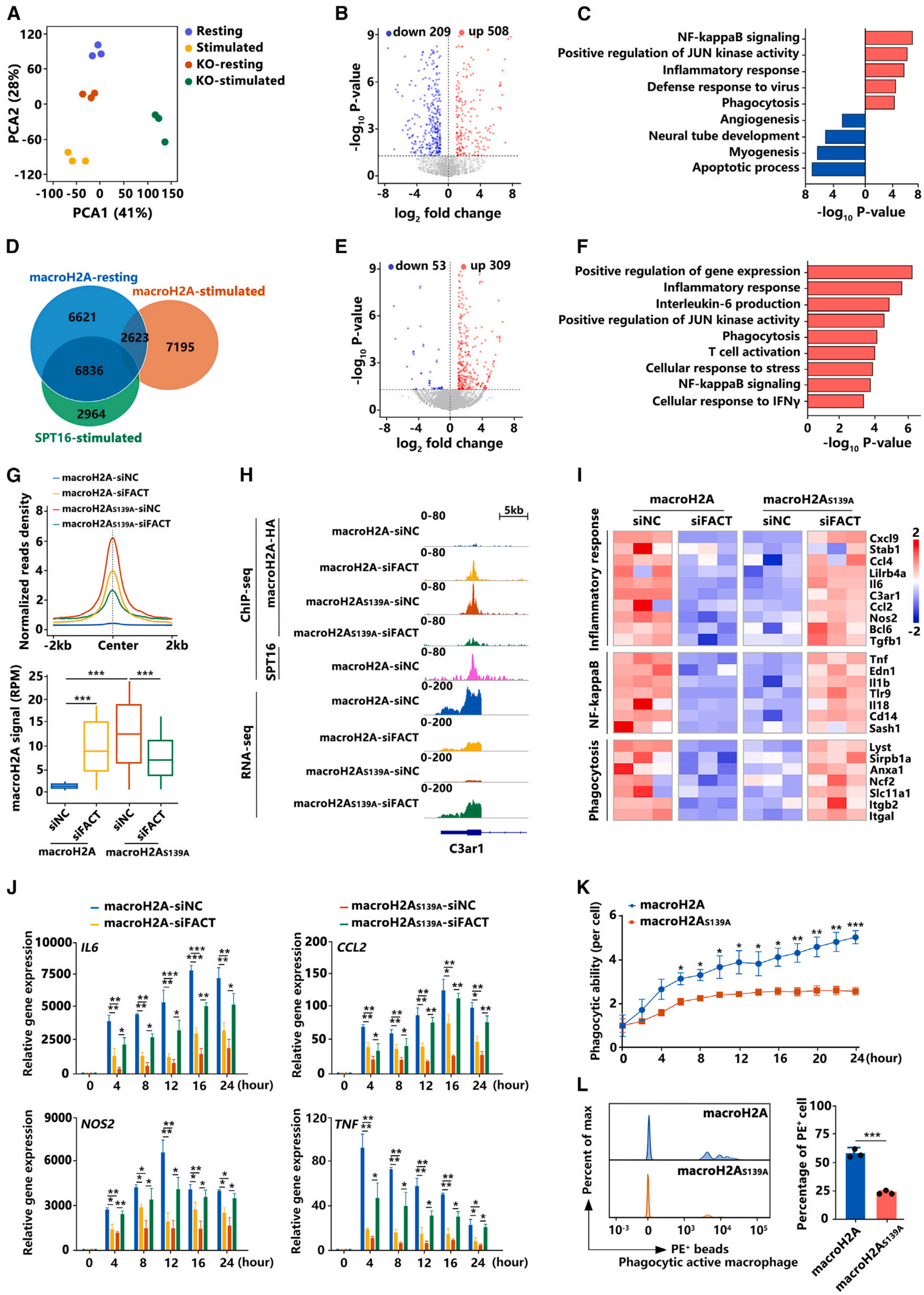
(N) Western blot analysis with anti-macroH2A primary antibodies on the purified endogenous macroH2A wild-type (WT) and macroH2A$_{S139A}$ from the RAW264.7 cell line separated by Phos-tag polyacrylamide gel to identify the phosphorylated macroH2A in their corresponding cells. The stripe within the red border was cut to detect individually with longer exposure time (bottom).

(O) Venn diagram analysis on the ChIP-seq data for macroH2A and SPT16 in macroH2A cells, macroH2A$_{S139E}$ in macroH2A$_{S139E}$ cells.

(P and Q) Density plot (P) and boxplots (Q) analysis on the reads density of macroH2A or macroH2A$_{S139E}$ at 5,025 peak regions in their corresponding cells. The data of (H) and (Q) were determined by Wilcoxon signed-rank test.

(M) was determined by two-tailed t test. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Negative-control short interfering (si) RNA (siNC); siRNA that targets SPT16 (siFACT).



(legend on next page)

macroH2A$_{S139A}$, the knockdown of FACT does not affect the distribution of macroH2A and the mutant S139 (Figure S3M). For the 6,203 regions in Figure 3F, the non-FACT-bound peaks gained in the S139A mutant, we found that the gained localizations of macroH2A$_{S139A}$ at these regions suppressed the expression level of the genes (Figure S3N), and the knockdown of FACT does not affect the distribution the macroH2A$_{S139A}$ (Figure S3O) and the expression level of the genes at these regions (Figure S3P). The results further confirmed that the knockdown of FACT only affects the signal of macroH2A and the mutant S139 at FACT-bound regions but does not affects the signal at non-FACT-bound regions.

As we revealed that the depletion or maintenance of macroH2A by the FACT complex may be precisely regulated by dynamic phosphorylation at the key residue S139, we further investigated whether S139 can be phosphorylated in macrophages. We purified the endogenous wild-type macroH2A and macroH2A$_{S139A}$ from the RAW264.7 cell lines and identified the S139-specific phosphorylation bands by the gel of Mn^{2+}-Phos-tag SDS-PAGE, which is used to analyze phosphoproteins.[42,43] As shown in Figure 3N, we can observe the band specifically correlated to macroH2A S139 phosphorylation, which disappears in macroH2A$_{S139A}$. We then constructed the phosphomimic mutant macroH2A$_{S139E}$ RAW264.7 cell line (Figure S3Q) and conducted ChIP-seq analysis to discern whether the phosphomimic mutant leads to the maintenance of macroH2A domains by FACT. ChIP-seq analysis revealed that 5,025 more macroH2A$_{S139E}$-bound regions are colocalized with FACT than in the wild-type macroH2A (Figure 3O), which indicates that in these 5,025 regions, the enriched FACT complex can deplete the wild-type macroH2A but cannot deplete the phosphomimic mutant macroH2A$_{S139E}$. Knocking down the endogenous FACT expression in the cells will cause massive enrichment of macroH2A at these sites—and the massive loss of macroH2A$_{S139E}$ (Figures 3P and 3Q). The transwell assay showed that, compared with the wild-type cells, migration abilities were significantly reduced in macroH2A cells with FACT knockdown and in macroH2A$_{S139E}$ cells, but rescued in macroH2A$_{S139E}$ cells with FACT knockdown (Figure S3R). The results further confirmed that the phosphomimic mutant at S139 residue leads to the switch of FACT's function *in vivo*.

FACT-mediated depletion of the macroh2a-nucleosome regulates the stimulation-induced transcription and cellular response of macrophages

We further test how FACT-mediated depletion of macroH2A-nucleosomes functions to regulate gene expression in a stimulation-dependent manner during macrophage activation. Western blot (Figure S4A) and time course quantitative PCR with reverse transcription (RT-qPCR) analysis (Figure S4B) confirmed the stimulation-induced nature of the cells. By comparing the RNA-seq data of the stimulated state for macroH2A-KO cells and macroH2A cells (Figure 4A), we found that a significant number of genes that functionally correlated to the immune response of macrophages, including transcription, "immune system process," "nuclear factor κB (NF-κB) signaling," and "phagocytosis," as revealed by GO analysis, are regulated directly or indirectly by macroH2A (Figures 4B, 4C, S4C, and S4D). The results indicated a dedicated role of macroH2A in stimulation-responsive transcription during macrophage activation. Further ChIP-seq analysis of macroH2A for the resting and stimulated macroH2A cells revealed a genome-widely dynamic re-localization of macroH2A during macrophage activation (Figure 4D). More than 85% of macroH2A-localized regions are depleted and repositioned into the 7,195 regions during the activation process. Most of the genes in the 7,195 regions with gained macroH2A upon stimulation are suppressed, with the expression level downregulated (Figure S4E). The position of FACT also displayed a genome-wide dynamic re-localization during macrophage activation (Figure S4F). Interestingly, in the stimulated state, almost none of macroH2A was detected to colocalize with FACT (Figure 4D, by comparing macroH2A-stimulated with SPT16-stimulated). More than 50% of macroH2A binding sites (6,836 regions) in the resting state will be bound by FACT and depleted during the activation process, by comparing the genome-wide localization of macroH2A in resting state (macroH2A-resting) versus the binding sites of FACT in the activation state (SPT16-stimulated) in Figure 4D. By coupling the 6,836 regions with RNA-seq analysis on the transcriptomic profile of the activation process, we defined 309 upregulated genes (Figure 4E), most of which are functionally associated with the immune response of macrophages and reflected their stimulation-induced nature, including

Figure 4. FACT-mediated depletion of macroH2A-nucleosomes regulates the stimulation-induced transcription and cellular response of macrophages

(A) PCA of macroH2A resting cells (resting) or stimulated cells (stimulated), macroH2A-KO resting cells (KO-resting) or stimulated cells (KO-stimulated), $n = 3$.
(B) Volcano plot of transcriptomic changes for macroH2A-bound genes identified by macroH2A ChIP-seq analysis between the macroH2A-KO and macroH2A stimulated cells.
(C) GO analysis for the upregulated (red) and downregulated (blue) genes defined in (B) by DAVID.
(D) Venn diagram analysis for the ChIP-seq data of macroH2A in macroH2A resting (macroH2A-resting) or stimulated (macroH2A-stimulated) cells and SPT16 in macroH2A stimulated cells (SPT16- stimulated).
(E) Volcano plot of transcriptomic changes for 6,836 peak-region-regulated genes between macroH2A-stimulated and resting cells.
(F) GO analysis for the upregulated genes defined in (E) by DAVID.
(G) Density plot (top) and boxplots (bottom) analysis on the reads density of macroH2A at 6,836 peak regions in their corresponding stimulated cells.
(H) The ChIP-seq tracks (top) and the RNA-seq tracks (bottom) at specific genomic regions, which are bound with macroH2A or SPT16 in their corresponding stimulated cells.
(I) Heatmap of GO terms defined in (F). Expression values are normalized row-wise as *Z* scores.
(J) RT-qPCR analysis on the genes *IL-6*, *CCL2*, *NOS2*, and *TNF* in their corresponding stimulated cells, $n = 3$.
(K) Statistical analysis of phagocytosis was measured using the IncuCyte S3 live cells analysis system in macroH2A and macroH2A$_{S139A}$ cells, $n = 3$.
(L) Quantification of phagocytosis was measured by flow cytometry in macroH2A and macroH2AS139A cells (left) and statistical analysis (right), $n = 3$. The data of (G) were determined by Wilcoxon signed-rank test.
(J)–(L) were determined by two-tailed t test. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

"inflammatory response," phagocytosis, and NF-κB signaling, as revealed by GO terms analysis (Figure 4F).

To further identify the correlation of FACT and macroH2A in the 6,836 regions, we knocked down the endogenous FACT expression and observed the localizations of macroH2A at these regions. We found that knockdown of FACT in macroH2A cells caused a massive enrichment of macroH2A at these sites in the activation state (yellow line in Figures 4G and 4H). For the mutant macroH2A$_{S139A}$, which reverses FACT-dependent depletion of macroH2A, ChIP-seq analysis revealed that the enriched FACT complex cannot deplete macroH2A$_{S139A}$ during the activation process, leaving the 6,836 FACT-bound regions colocalized with macroH2A$_{S139A}$ (orange line in Figures 4G and 4H), while knockdown of FACT in the cells causes a massive loss of macroH2A$_{S139A}$ at these sites in the activation state (green line in Figures 4G and 4H). The genome-wide analysis is consistent with our *in vitro* investigations, which found that FACT functions to deplete macroH2A while the S139A mutation in macroH2A reverse this function.

The non-depletion or mis-localization of macroH2A due to the macroH2A$_{S139A}$ mutation or knockdown of FACT greatly impairs the stimulation-induced transcription and cellular response during macrophage activation. RNA-seq analysis revealed that the gene expression upregulated by FACT-mediated macroH2A depletion during activation (Figure 4I, macroH2A-siNC) is repressed by knockdown of FACT (Figure 4I, macroH2A-siFACT) or macroH2A$_{S139A}$ mutation (Figure 4I, macroH2A$_{S139A}$-siNC). The depression effect of macroH2A$_{S139A}$ can be rescued by the knockdown of FACT (Figure 4I, macroH2A$_{S139A}$-siFACT). RT-qPCR measurement of the expression levels for the typical genes revealed by RNA-seq analysis, including *IL6*, *CCL2*, *NOS2*, and *TNF* (Figure 4J), revealed (1) the reduced induction in macroH2A$_{S139A}$ cells, (2) the reduced induction in macroH2A cells with FACT knockdown, and (3) the potent rescued induction in macroH2A$_{S139A}$ cells with FACT knockdown. We also investigated the phagocytic ability of the stimulated cells. Remarkably, the mutant macroH2A$_{S139A}$ greatly impairs the ability to phagocytose particles as compared with macroH2A cells (Figures 4K and 4L). The results revealed that FACT-mediated depletion of macroH2A-nucleosomes regulates the stimulation-induced transcription and cellular response of macrophages (Figure 5).

DISCUSSION

Although macroH2A is generally associated with condensed chromatin and transcriptionally inactive genes, its precise role in regulating the chromatin state and function in distinct cell types, tissues, and disease contexts remain elusive. Our study provides mechanistic insights into macroH2A's regulatory functions in nucleosome dynamics and its direct impact on the function of the FACT complex. We demonstrated that macroH2A, regardless of whether its macro or linker domain is truncated or not, does not affect the mechanical stability or folding kinetics of the nucleosome. We revealed that, although the linker domain of macroH2A does not affect the mechanical stability or folding kinetics of the nucleosome, it plays a key role in blocking the nucleosome maintenance's function of FACT. Given that the amino acid composition of the linker domain of macroH2A is similar to the C terminus of histone H1, previous *in vitro* investigations have shown that the linker domain can increase the protection of the nucleosome entry/exit DNA site from exonuclease III digestion, reduce the availability of extra-nucleosomal DNA for Chd1 binding,[10] and function to promote the oligomerization and condensation of chromatin fibers in the absence of the macro domain.[11] In addition, the linker domain has been revealed to limit the chromatin plasticity and function to stabilize heterochromatin architecture in living cells.[44] Our previous investigations by magnetic tweezers revealed that the linker histone H1 enhances the stability of the nucleosome by locking the outer DNA wrap of the nucleosome, with the unfolding force rising from ~5 pN for nucleosome without H1 to ~10 pN for nucleosomes bound with H1 at the chromatin state.[35] However, for the macroH2A, our investigations showed that whether the linker domain is truncated or not, it does not affect the nucleosome stability. This finding refines our understanding of macroH2A's role in chromatin structure. Although the linker domain does not affect the mechanical stability of the nucleosome—especially the strength of interactions between the DNA and histones—it may affect nuclease digestion and the function or access of other proteins near the dyad DNA at the nucleosome.

Intriguingly, we discovered that macroH2A incorporation significantly impairs FACT's function in nucleosome maintenance. The histone chaperone FACT has been found to play essential roles in almost all chromatin-related processes, including DNA transcription, replication, and repair.[22,23] Previous studies have revealed the dual functions of FACT, including not only destabilizing the nucleosome and facilitating the progression of DNA and RNA polymerases on chromatins but also maintaining the genome-wide integrity of chromatin structure.[24–30] Recent structural and single-molecular studies showed that the FACT complex can tether both the $(H3\text{-}H4)_2$ tetramer and H2A-H2B dimer on DNA to form a loose but undetached nucleosome state to fulfill these apparent opposite functions.[31,32] It is of critical importance to investigate how the core functions of FACT are regulated by different epigenetic factors to balance FACT's function in different circumstance. Our previous studies have revealed the regulation of H2A and H2B mono-ubiquitination on the functioning of FACT.[36,45] H2AK119ub has been found to inhibit the binding of FACT on nucleosomes and to block FACT's function at the nucleosome level,[36] while H2BK120ub enhances the binding of FACT and greatly impairs the destabilized effect of FACT at nucleosome state.[45] In addition, FACT has been found to be required for the pruning of macroH2A from the transcribed chromatin region to establish and maintain the genomic localization of macroH2A variants.[33] However, how macroH2A directly regulates the function of FACT remains unclear. In this paper, using *in vitro* single-molecular magnetic tweezers, we dissected FACT's function in mediating the depletion of the macroH2A-nucleosome. Furthermore, we identify the residue S139 in the linker region of macroH2A as the key site to switch FACT's function in nucleosome maintenance/depletion. Our research provides mechanistic insight into the distinct regulation of macroH2A on FACT's functions and helps propose a unifying model for understanding the interplay of macroH2A and FACT's activity under different chromatin circumstances—not only in transcription but also in DNA damage repair and replication processes.

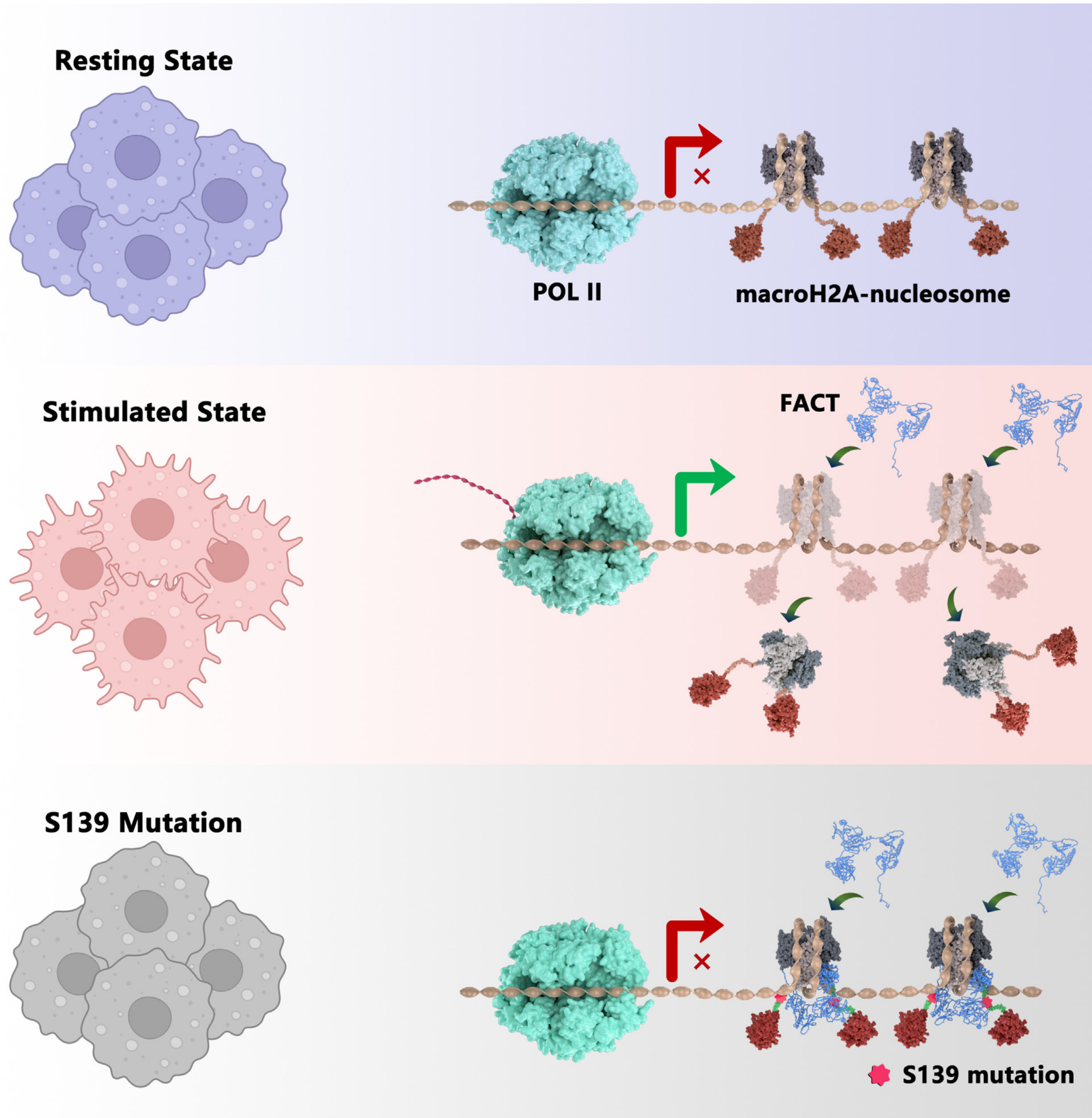


Figure 5. FACT-mediated depletion of macroH2A-nucleosomes allows the correct localization of macroH2A and proper stimulation-induced transcription in macrophages

Stimulation of macrophages with LPS and IFNγ causes huge genome-wide repositioning of macroH2A, where FACT-mediated depletion of macroH2A-nucleosomes plays an essential role to regulate the cellular response. S139 mutation in macroH2A, which reverses the depletion function of FACT on the macroH2A-nucleosome, reshapes the position of macroH2A and mis-regulates the cellular states of macrophages.

As a highly conserved and widespread histone variant in vertebrates, macroH2A has been shown to function as a tumor suppressor in a wide range of cancers and play a critical role in stabilizing cell identity by posing a barrier to somatic cell reprogramming toward pluripotency. Most recently, Filipescu et al. revealed that macroH2A deficiency enhances inflammatory gene expression, leading to an impaired anti-tumor immune response with accumulation of immunosuppressive monocytes and depletion of functional cytotoxic T cells in a macroH2A-deficient melanoma mouse model.[41] However, how macroH2A functions in the immune response of macrophages remains unclear. In this paper, we found that macroH2A plays a critically important role in the dynamic regulation of macrophage cellular states (Figure 5). KO of macroH2A greatly enhances the cellular response of macrophages. Furthermore, stimulation of macrophages with lipopolysacchariade (LPS) and interferon (IFN)γ causes significant genome-wide repositioning of macroH2A, where FACT-mediated depletion of macroH2A-nucleosomes plays an essential role in regulating the cellular response. S139 mutation in macroH2A, which reverses the function of FACT on the macroH2A-nucleosome, reshapes the position of macroH2A and regulates stimulation-induced transcription. The requirement for FACT-mediated depletion of macroH2A is suggested to permit the re-localization of macroH2A in chromatin, allowing rapid and proper stimulation-induced transcription to inflammatory stimuli and regulating the diverse immune responses of macrophages to environments and diseases states. Our investigations provide mechanistic insights into how the histone variant macroH2A co-regulates with FACT and functions in the cellular response of macrophages in immunity (Figure 5). In addition, the phosphorylation of macroH2A at S139 has been observed by mass spectrometry in UV-damaged M059K glioblastoma cells,[40] but its function remains unclear. Our investigations showed that macroH2A, bearing the phosphor-mimicking glutamic acid mutation at S139 (S139E). can reverse the depletion function of FACT on macroH2A-nucleosomes *in vitro* and reshapes the position of macroH2A to regulate the stimulation-induced transcription process in macrophage cells. The results suggest that the depletion or maintenance of macroH2A by the FACT complex may be precisely regulated by dynamic modification at the key residue S139 *in vivo*. Our investigations show that macroH2A, bearing the phosphor-mimicking glutamic acid mutation at S139 (S139E), can reverse the depletion function of FACT on macroH2A-nucleosomes *in vitro* and reshape the position of macroH2A to regulate the stimulation-induced transcription process in macrophage cells.

Limitations of the study

The histone chaperone FACT plays a crucial role in a wide range of chromatin-related activities. However, the specific impact of macroH2A on FACT's functions remains to be defined. Our research dissects how FACT mediates the depletion of the macroH2A-nucleosome and pinpoints the pivotal S139 residue in macroH2A's linker region, which modulates FACT's role in maintaining or depleting nucleosomes. Our study utilizes the homotypic macroH2A-nucleosomes; further exploring FACT's role in heterotypic nucleosomes that contain both macroH2A and H2A could yield valuable insights. In addition, we establish that FACT-mediated depletion of macroH2A facilitates its proper genome-wide distribution. The S139 mutation, including the phosphor-mimetic S139E variant, alters macroH2A distribution and affects stimulus-induced transcription and the cellular response in macrophages. Given the previous detection of macroH2A S139 phosphorylation in UV-damaged M059K glioblastoma cells via mass spectrometry, future research should elucidate the biological significance and regulatory mechanism of S139 phosphorylation in modulating FACT's function *in vivo*.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.molcel.2024.07.011.

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AUTHOR CONTRIBUTIONS

P.C. and W.L. conceived and supervised the project. D.J. performed cell biology experiments, the ChIP-seq and RNA-seq experiments, and genome-wide data analysis. X.X. performed the magnetic tweezers and AFM analysis. A.L. prepared the proteins and nucleosome samples and performed the biochemical analysis. X.F., J.M., and D.W. helped to prepare the proteins and cell lines. M.X., L.M., and P.-Y.W. helped to discuss the project. D.J., W.L., and P.C. analyzed the data and wrote the manuscript with input from all co-authors.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
anti-HA	Abcam	Cat#ab9110; RRID:AB_307019
anti-H2A	Abcam	Cat#ab18255; RRID:AB_470265
anti-SPT16	Cell Signaling Technology	Cat#12191; RRID:AB_2732025
anti-macroH2A1.2	Cell Signaling Technology	Cat#4827; RRID:AB_1950388
anti-SSRP1	Biolegend	Cat#609702; RRID:AB_315731
anti-Spike-in	Active Motif	Cat#61686; RRID:AB_2737370
anti-actin	ABclonal	Cat#AC026; RRID:AB_2768234
anti-H3	Abclonal	Cat#A2348
goat anti-rabbit second antibody	Zhongshan Golden Bridge	ZB2301
goat anti-mouse second	Zhongshan Golden Bridge	ZB-2305
Bacterial and virus strains		
Trans1-T1	TransGen Biotech	CD501-02
BL21	TransGen Biotech	CD601-02
Trans5α	TransGen Biotech	CD201-01
Chemicals, peptides, and recombinant proteins		
DMEM medium	Gibco	C11995500BT
Fetal Bovine Serum	Hyclone	SH30071
MNase	NEB	M0247S
Protein A/G-Dynabeads	Thermo Fisher Scientific	10015D
IFNγ	BioLegend	575304
Lipopolysaccharide	InvivoGen	tlrl-3pelps
Zeosin	Gibco	R25001
Puromycin	InvivoGen	ant-pr-1
Lipofectamine 3000	Thermo Fisher Scientific	L3000015
anti-Flag M2-agarose	Sigma	A2220
Flag peptide	Sigma	F4799
Streptavidin-agarose-beads	Thermo Fisher Scientific	29200
2.8-μm diameter Dynabeads	Invitrogen Norway	M280
RNA Extraction Reagent	Vazyme	R401-01
Reverse transcribed reagent Kit	Vazyme	R223-01
SYBR qPCR Master Mix	Vazyme	Q711-03
NeutrAvidin agarose resin	Thermo Fisher Scientific	29200
PVDF membrane	Millipore	ISEQ00010
Pierce Western ECL substrate	Thermo Fisher Scientific	32109
Invasion chamber	Corning	3422
PE-labeled beads	Thermo Fisher Scientific	F13083
Deposited data		
Original code	This paper	Mendeley Data: https://doi.org/10.17632/93kzdwvkyr.1
Unprocessed gel images data	This paper	Mendeley Data: https://doi.org/10.17632/883bp89my8.1
RAW264.7 RNA-seq	This paper	GSE248744
RAW264.7 ChIP-seq	This paper	GSE248551

(*Continued on next page*)

Continued		
REAGENT or RESOURCE	SOURCE	IDENTIFIER
Experimental models: Cell lines		
RAW264.7 cells	This Paper	N/A
HEK 293T cells	This Paper	N/A
Oligonucleotides		
Primer:CCL2 forward:TTTTTGTCACCAAGCTCAAGAG	This Paper	N/A
Primer:CCL2 reverse:TTCTGATCTCATTTGGTTCCGA	This Paper	N/A
Primer:TNF Forward:ATGTCTCAGCCTCTTCTCATTC	This Paper	N/A
Primer:TNF reverse:GCTTGTCACTCGAATTTTGAGA	This Paper	N/A
Primer:SSRP1 forward:GCTCGTCATCCAAGTCTTCATCC	This Paper	N/A
Primer:SSRP1 reverse:CTCGCTCCGCCTCCTCTTC	This Paper	N/A
Primer:SPT16 reverse:ATCAATGCTATTCCTGTTGCCTCTC	This Paper	N/A
Primer:SPT16 reverse:CCGTCCTCAGCGTCACTCC	This Paper	N/A
Primer:IL-6 reverse:CTCCCAACAGACCTGTCTATAC	This Paper	N/A
Primer:IL-6 Reverse:CCATTGCACAACTCTTTTCTCA	This Paper	N/A
Primer:NOS2 forward:ATCTTGGAGCGAGTTGTGGATTGTC	This Paper	N/A
Primer:NOS2 reverse:TAGGTGAGGGCTTGGCTGAGTG	This Paper	N/A
Recombinant DNA		
Wild-type human macroH2A1.2	This Paper	N/A
Mutated human macroH2A$_{S139A}$	This Paper	N/A
Mutated human macroH2A$_{S139E}$	This Paper	N/A
Mutated human macroH2A$_{S137E}$	This Paper	N/A
Mutated human macroH2A$_{S139A-Q140A}$	This Paper	N/A
Truncated human macroH2A$_{160}$	This Paper	N/A
Truncated human macroH2A$_{143}$	This Paper	N/A
Truncated human macroH2A$_{136}$	This Paper	N/A
Truncated human macroH2A$_{121}$	This Paper	N/A
FACT complex	This Paper	N/A
Software and algorithms		
Bowtie2	Langmead and Salzberg[46]	http://bowtie-bio.sourceforge.net/bowtie2/other_tools.shtml
Samtools	Li et al.[47]	http://samtools.sourceforge.net
MACS	Zhang et al.[48]	https://github.com/macs3-project/MACS
R	https://www.r-project.org/	https://www.r-project.org/
Hisat2	Kim et al.[49]	http://daehwankimlab.github.io/hisat2
DESeq2	Love et al.[50]	https://bioconductor.org/packages/3.17/bioc/html/DESeq2.html
DAVID	https://david.ncifcrf.gov/	https://david.ncifcrf.gov/
Graphpad Prism	GraphPad software	https://www.graphpad.com
ImageJ	https://imagej.nih.gov/ij/	https://imagej.nih.gov/ij/

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Ping Chen (chenping@ccmu.edu.cn).

Materials availability

Plasmids generated in this study will be made available on request, but we may require a payment and/or a completed Materials Transfer Agreement if there is potential for commercial application.

Data and code availability

- All the sequencing data generated in this study have been deposited to Gene Expression Omnibus (GEO) database under the accession number GSE248551 and GSE248744. All unprocessed images generated in this study have been deposited to Mendeley Data (https://doi.org/10.17632/883bp89my8.1). These data are publicly available as of the date of publication.
- All original code generated in this study have been deposited to Mendeley Data (https://doi.org/10.17632/93kzdwvkyr.1).
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Cell culture

The RAW264.7 cells were cultured in DMEM (Gibco, C11995500BT) with 10% fetal bovine serum (FBS, Hyclone, SH30071), 1% penicillin and 1% streptomycin at 37°C in 5% CO_2. For macrophage activation, RAW264.7 cells were incubated with 20 ng/mL recombinant mouse IFNγ (BioLegend, 575304) and 100 ng/mL lipopolysaccharide (InvivoGen, tlrl-3pelps) for 0-24 hours.

MacroH2A1, which is encoded by H2afy, knockout cell line was generated by CRISPR/Cas9-mediated genome editing using pX260 vector (Table S1). The guide sequence for the macroH2A is: 5′-GCGGGGGCGTGTTGCCGAAT-3′. After transfection, the cells were plated into 10 cm culture plates and screened with 100 μg/mL zeosin (Gibco, R25001). Single-cell clones were picked out, with the genotypes identified by PCR-coupled DNA sequencing and western blot.

HEK 293T cells were cultured at 37°C in DMEM with 10% FBS under 5% CO_2 conditions and used to package lentivirus vectors. HEK 293T cells were transfected with constructs containing cDNA of macroH2A1.2 in pLVX vector (Table S1), including the wild-type macroH2A and the mutant macroH2A$_{S139A}$. After 48 hours' transfection, the culture medium was collected to harvest lentivirus particles. RAW264.7 macroH2A-KO cells were infected with lentivirus, and stable cell line were generated by 1 μg/mL puromycin selection (InvivoGen, ant-pr-1). The overexpression of macroH2A was evaluated by western blot. Cell cultures were periodically tested for mycoplasma contamination.

For SPT16 knockdown in RAW264.7 cells, the sequence of SPT16 siRNA oilgos is: sense 5′-GCAUCACCUCUGAGGUCUUTT-3′,antisense 5′-AAGACCUCAGAGGUGAUGCTT-3′. SiRNA was transfected into RAW264.7 cells with Lipofectamine 3000 (Thermo Fisher Scientific, L3000015) according to the manufacturer's protocol. After 48 hours' transfection, cells are processed and used for subsequent experiments.

METHOD DETAILS

Protein purification

The wild-type or mutated human macroH2A1.2 was co-expressed in bacteria with H2B in a pRSF-duet histone expression vector. The plasmid was transformed into *Escherichia coli* BL21 (DE3), grown to the cell density reached optical density at 600 nm values between 0.5 and 0.6 at 37°C, and induced 16 hours at 16°C with the addition of 0.4 mM isopropylthio-β-galactoside (IPTG). MacroH2A-H2B dimer was purified over HiTrap-SP HP column (GE Healthcare) followed by gel filtration over a Superdex 200 column (GE Healthcare) in the presence of 2 M NaCl. Histones H2A, H2B, H3, H4, macroH2A$_{121}$ and macroH2A$_{160}$ were cloned into pET3a histone expression plasmid, overexpressed in *Escherichia coli*, and purified from inclusion bodies.[51] For the purification of recombinant FACT complex,[23,32] Sf9 cells (1.5-2 × 10^6 /mL) were infected with baculovirus containing Flag-SPT16 and His$_6$-SSRP1, and incubated at 27°C for 72 hours. The infected cells were collected and washed with ice-cold PBS, and lysed in Flag-150 buffer (150 mM NaCl, 20 mM Tris-HCl, pH 8.0, 5% glycerol, 1 mM PMSF). 500 μl anti-Flag M2-agarose (Sigma, A2220) were incubated with 60 mL (10 mg) of the cell extracts for 4 hours at 4°C, and then washed by Flag-150 buffer. The bound FACT proteins were eluted in 0.5 mg/mL Flag peptide (Sigma, F4799), and purified by a Heparin HP column (GE Healthcare). The purified FACT complex was then dialyzed against BC-100 buffer (100 mM NaCl, 10 mM Tris-HCl, pH 8.0, 0.5 mM EDTA, 20% glycerol, 1 mM DTT, 1 mM PMSF) and stored at -80°C.

Nucleosome or tetrasome reconstitution

To reconstitute the respective canonical histone octamers, H3-H4 tetramers and H2A-H2B dimers,[32] equimolar amounts of individual histones were mixed in unfolding buffer (7 M guanidinium HCl, 20 mM Tris-HCl, pH 7.5, 10 mM DTT), dialyzed into 2 M NaCl refolding buffer (2 M NaCl, 10 mM Tris-HCl, pH 7.5, 1 mM EDTA, 5 mM 2-mercaptoethanol), and further purified by a Superdex S200 column. To reconstitute the wild-type and mutated macroH2A octamers, H3-H4 tetramers and two-fold molar ratio of macroH2A-H2B dimers were mixed and dialyzed into 2.4 M NaCl refolding buffer (2.4 M NaCl, 10 mM Tris-HCl, pH 7.5, 1 mM EDTA, 5 mM 2-mercaptoethanol), and purified by a Superdex S200 column. For magnetic tweezers' investigation, the 409-base pair (bp) DNA template containing a single "601" positioning sequence in the middle was prepared by PCR using a biotin-labeled forward primer and a 3 × digoxigenin (3 × dig)-labeled reverse primer. For mono-nucleosome pull-down assay, the biotin-labeled 208 bp "601" middle positioning DNA was prepared by PCR with biotin-labeled primer. Nucleosome and tetrasome samples were assembled using the salt-dialysis method. The reconstitution reaction mixture with octamers or tetramer and 601 based DNA templates were mixed in TEN buffers (10 mM Tris-HCl, pH 8.0, 1 mM EDTA, 2 M NaCl), and dialyzed over 16 hours at 4°C in TEN buffer with the concentration of NaCl continuously diluted from 2 M to 0.6 M by slowly pumping in TE buffer (10 mM Tris-HCl, pH 8.0, 1 mM EDTA), and then followed by a final dialysis step in TE buffer for 4 hours. The stoichiometry of the octamer or tetramer to the DNA template was determined by agarose gel electrophoresis and AFM analysis.

Mono-nucleosome pull-down assay

10 μl of the streptavidin-agarose-beads (Thermo Fisher Scientific, 29200) for each reaction was washed by PBS 3 times and 5 min each time, and then washed by BC300 buffer (20 mM Tris-HCl, pH 7.5, 10% glycerol, 300 mM NaCl, 0.1% NP-40, 500 μg/mL BSA) 3 times at 4°C. 1 μg Biotin-labeled mono-nucleosome were mixed with the beads in BC300 buffer, and incubated with increasing concentrations (4 nM and 8 nM) of the recombinant FACT complex overnight at 4°C. The beads were then washed by the buffer (20 mM Tris-HCl, pH 7.5, 10% glycerol, 300 mM NaCl, 0.5% NP-40) for 3 times, 10 min each time. The proteins bound on the beads were eluted with SDS-loading buffer (50 mM Tris-HCl, pH 6.8, 2% SDS, 0.1% Bromophenol blue, 10% Glycerol, 1% beta-mercaptoethanol), and analyzed by western blot. The uncropped and unprocessed scans of all the blots shown in the Source Data file.

Flag-IP assay

For the Flag-IP assay, 10 μl of the anti-Flag M2-agarose (Sigma, A2220) beads for reaction was washed by PBS 3 times, and then precleaned in BC300 buffer for 1 hour at 4°C. The recombinant Flag-FACT complex (8 nM) and increasing concentrations (1.5 nM and 3 nM) of H2A-H2B dimer, macroH2A-H2B dimer and macroH2A$_{S139A}$-H2B dimer were mixed with the beads, and incubated overnight at 4°C.[52] The beads were washed by the buffer (20 mM Tris-HCl, pH 7.5, 10% glycerol, 300 mM NaCl, 0.5% NP-40) 3 times, 10 min each time. Finally, the proteins bound on the beads were eluted by SDS-loading buffer and analyzed by western blot. The uncropped and unprocessed scans of all of the blots shown in the Source Data file.

AFM analysis

20 μl of spermidine (10 mM) was dropped onto newly cleaved mica surface and incubated for 10 min. Then the mica was rinsed with 200 μl ddH_2O for 4 times, with the surface blowed with nitrogen gas to dry briefly. 10 μl of the nucleosome samples (1 ng/μl) was added onto the mica surface, incubated for 10 to 15 min. Finally, the mica was washed with 200 μl ddH_2O for 4 times and blowed to dry gently. The prepared AFM samples were examined using ScanAsyst Mode of AFM (MultiMode 8 SPM system, BRUKER).

Single-molecule magnetic tweezers analysis

The single-molecule force-extension measurements were performed by magnetic tweezers.[36] To stretch the reconstituted mono-nucleosome, the two DNA ends were separately tethered to a glass coverslip by digoxigenin-anti-digoxigenin ligation and to a 2.8-μm diameter Dynabeads (M280, Invitrogen Norway) by biotin-streptavidin ligation, respectively. The tensions applied on the super paramagnetic Dynabeads were raised from the strong magnetic field gradient of the two small NdFeB magnets. The magnets were controlled to pull the Dynabeads and stretch the DNA molecule on nucleosome. To record the real-time position (x, y, z) of the Dynabead at various forces, the images of the bead were projected onto the CCD camera (MC1362, Mikrotron) through an inverted microscope objective (UPLXAPO60XO, NA 1.42, Olympus), with the diffraction pattern of the bead calculated and compared at various distances from the focal point of the objective. The three-dimensional position of the beads in the flow cell were traced by using the quadrant-interpolation (QI) algorithm applied by LabVIEW software.[53] The forces applied to the beads were calculated by a 10,000 bp DNA tethered between the bead (M280) and the coverslip and repeated for 10 independent DNA molecules. At each magnet position, the y-position of the bead was recorded at 500 Hz for 5 min with the corresponding force calibrated by power-spectral-density (PSD) analysis. The relationship between force and magnet position was fitted well with a double exponential function, and used to calculate the forces applied in force-extension measurements of nucleosome.

Force-extension measurement of mono-nucleosome

A flow cells consists of a bottom functionalized coverslip, a double-sided tape with a rectangular channel (5 × 50 mm^2), and an upper coverslip with two holes at each end to be used as inlet and outlet. The flow cell was first incubated with anti-digoxigenin (0.1 mg/mL) for at least 1 hour and then passivated with 10 mg/mL BSA for at least 2 hours. Nucleosome samples labeled with digoxigenin and biotin at each end of DNA were then injected into the flow cells and tethered between the super-paramagnetic bead (M280 Invitrogen Norway) and the bottom coverslip. To trace the detail of the structural transition of nucleosome samples, magnets were moved near to the samples at a rate of 10 μm/s from 0 pN to 30 pN, during which the dynamic unfolding of the nucleosome was recorded. After that, force was reduced to 0.1 pN rapidly and waited for 5 min, and then repeated the stretching cycle on the same nucleosome at least 3 times. The force-extension curves and statistical measurements were processed using LabVIEW software and MATLAB. In addition to studying the force-extension curves of nucleosome samples, we calculated the probability of nucleosome maintenance through three stretches. For the H2A nucleosome experiments involving FACT, we conducted three separate batches (the nucleosome number we examined in each batch is 30, 29 and 43). In each batch, every individual molecule was stretched three times. The first stretch consistently displayed a nucleosome signal, maintaining a 100% nucleosome signal ratio. During the second stretch, the proportion of nucleosome signals decreased to 96%, 93%, and 97.7% (the numbers are 29, 27 and 42, accordingly) across the three batches, averaging 96.1%. The nucleosome signal retention further declined during the third stretch to 80%, 82.8%, and 90.7% (the numbers are 24, 24 and 39, accordingly), with an average of 85.3%. In contrast, for the macroH2A nucleosome experiments with FACT, the nucleosome signal retention during the first stretch in all three batches remained at 100% (the nucleosome number we examined in each batch is 34, 40 and 32). However, during the second stretch, the proportions significantly dropped to 5.9%, 7.5%, and 6.3% (the numbers are 2, 3 and 2, accordingly), with an average of 6.6%. The retention of nucleosome signals during the third stretch further decreased to 5.9%, 2.5%, and 0%, (the numbers are 2, 1 and 0, accordingly) resulting in an average of 2.96%.

Real-time deposition assay by single-molecule magnetic tweezers

The real-time deposition process of H2A-H2B or macroH2A-H2B dimer onto the reconstituted tetrasome in the presence of FACT were traced by magnetic tweezers, respectively. The mono-$(H3\text{-}H4)_2$ tetrasome were anchored with magnetic tweezers, as we described for the mono-nucleosome. 100 μl of the reconstituted FACT complex (50 nM) were injected with H2A–H2B or macroH2A-H2B dimer (50 nM) into the flow cells. The deposition process was traced immediately at 2.1 pN. Two sequential 10-nm deposition processes were recorded, which correspond to the formation of the two halves' outer DNA wrap, respectively. The force-extension measurement for the same reassembled sample were then carried out and recorded.

ChIP-seq analysis

The ChIP-seq assay was performed as previously described.[54] Briefly, for macrophage activation, RAW264.7 cells were incubated with 20 ng/mL IFNγ and 100 ng/mL lipopolysaccharide for 8 hours. The cells were crosslinked in adherent conditions with 1% formaldehyde for 15 min at room temperature and quenched with glycine to a final concentration of 125 nM. Fixed cells were lysed in NLB buffer (50 mM Tris-HCl, pH 8.0, 10 mM EDTA, 1% SDS, protease inhibitors) to extract nuclei, and sheared by sonication to 200-500-bp-sized fragments. Sheared chromatin was incubated with rabbit anti-HA antibodies (Abcam, ab9110, 3 μg) and anti-macroH2A antibodies (Cell Signaling Technology, 4827) and protein A/G-Dynabeads (Life Technology, 10 μl) for 12 hours at 4°C. The beads were washed with RIPA150 (50 mM Tris-HCl, pH 8.0, 150 mM NaCl, 1 mM EDTA, 0.1% SDS, 1% Triton X-100, 0.1% sodium deoxycholate), RIPA500 (50 mM Tris-HCl, pH 8.0, 0.5 M NaCl, 1 mM EDTA, 0.1% SDS, 1% Triton X-100, 0.1% sodium deoxycholate), RIPA-LiCl (50 mM Tris-HCl, pH 8.0, 1 mM EDTA, 1% NP-40, 0.7% sodium deoxycholate, 0.5 M LiCl) and TE buffer (10 mM Tris-HCl, pH 8.0, 1 mM EDTA), respectively. The beads were resuspended with NLB buffer. After de-crosslinked at 65°C for 6 hours, DNA fragments were digested with RNase A and proteinase K. Purified DNA fragments were generated libraries followed by high throughput sequencing using Illumina Novaseq6000. All ChIP are sequenced to generate paired-end data with the fragment length of 150bp. The antibodies used in this study included anti-SPT16 antibodies (Cell Signaling Technology, 12191, 3 μg), anti-Spike-in antibodies (Active Motif, 61686, 0.5 μg). All ChIP-seq data presented in the figures were repeated in two independent biological replicates. We check the repetition rate of two ChIP-seq data, and then select one of the duplicates for analysis. For the ChIP-seq data analysis, mus musculus (mm10) were mapped to all sequencing data by bowtie2[46] (version 2.2.5), with the default parameters. A fixed amount of reference epigenome (*Drosophila*) is added to each group. After mapping, the reads are normalized to the percentage of reference genome reads in the sample. Low quality reads were removed by samtools[47] (Version: 1.10) and only uniquely mapped reads were kept. The enriched regions (peaks) were detected using MACS[48] (version 1.4.2). HOMER was used to normalize the data, annotate the peaks and count the read density at peak regions.[55] UCSC Genome Browser was used to view the data tracks.[56] Boxplots were generated by R (http://www.r-project.org/).

RNA-seq analysis

RAW264.7 cells were cultured and treated as described in ChIP-seq analysis. The total RNA was extracted with RNA isolater Total RNA Extraction Reagent (Vazyme, R401-01). The RNA-seq was performed in the Novogene Co., Ltd (Beijing, China). The poly(A) RNA-seq libraries were prepared according to the Illumina TruSeq protocol and sequenced using Illumina Novaseq6000. The RNA-seq experiment was performed in three biological replicates. The pair-end RNA-seq reads were mapped to the Mus musculus (mm10) using hisat2[49] (version 2.1.0) with the default parameters, and the only uniquely mapped reads were used for further analysis (Table S2). The number of annotated genes is consistent in all groups, with a total of 55492 transcripts each group. Then, all annotated genes were used to generate the PCA. For coupling with the ChIP-seq and RNA-seq data, we define the macroH2A-bound genes overlapped with macroH2A ChIP-seq signals in the regions including the TSS, promoter, intron, and exon to conduct analysis.

The differential expression between samples was performed with DESeq2[50] (version 3.16). Gene enrichment was analyzed by the Database for Annotation, Visualization and Integrated Discovery (DAVID, https://david.ncifcrf.gov/).

RT-qPCR analysis

Total RNA was extracted from cells by RNA isolater Total RNA Extraction Reagent (Vazyme, R401-01). The reverse transcribed reagent Kit (Vazyme, R223-01) were used for cDNA synthesis. cDNA was amplified by real-time quantitative PCR in the Real-Time PCR system (ABI 7300, USA), using ChamQ Universal SYBR qPCR Master Mix (Vazyme, Q711-03). The primer pairs used for the qPCR experiments were the following:

CCL2: sense 5′-TTTTTGTCACCAAGCTCAAGAG-3′,
antisense 5′-TTCTGATCTCATTTGGTTCCGA-3′;
TNF: sense 5′-ATGTCTCAGCCTCTTCTCATTC-3′,
antisense 5′-GCTTGTCACTCGAATTTTGAGA-3′;
IL-6: sense 5′-CTCCCAACAGACCTGTCTATAC-3′,
antisense 5′-CCATTGCACAACTCTTTTCTCA-3′;
iNOS: sense 5′-ATCTTGGAGCGAGTTGTGGATTGTC-3′,
antisense 5′-TAGGTGAGGGCTTGGCTGAGTG-3′;
SSRP1: sense 5′-GCTCGTCATCCAAGTCTTCATCC-3′,
antisense 5′-CTCGCTCCGCCTCCTCTTC-3′;
SPT16: sense 5′-ATCAATGCTATTCCTGTTGCCTCTC-3′,
Antisense 5′-CCGTCCTCAGCGTCACTCC-3′.

All the results were calculated by three independent replications.

Mono-nucleosome IP assay

Briefly, the cells were lysed in hypotonic lysis buffer (10 mM tris-HCl, pH 7.9, 1.5 mM $MgCl_2$, 10 mM KCl, 0.1% Triton X-100, 0.5 mM PMSF) for 10 min at 4°C to extract nuclei. 0.5 mL MNase buffer (10 mM tris-HCl, pH 7.4, 20 mM KCl, 2 mM $CaCl_2$) were added to the resuspended nuclei and incubated for 10 min at 37°C. Then 3 μl Micrococcal Nuclease (NEB, M0247S) were added to digest the nuclei 12 min at 37°C to generate mono-nucleosome. The reaction was stopped by adding EGTA (15 mM). Mono-nucleosome were extracted with 500 μl buffer (10 mM tris-HCl, pH 7.9, 2 mM $MgCl_2$, 300 mM KCl, 0.1% Triton X-100) for 8 hours at 4°C. 3 μg anti-HA antibodies and NeutrAvidin agarose resin (10 μl, Thermo Fisher Scientific 29200) were mixed in the mono-nucleosome suspension for 8 hours at 4°C. Then the beads were washed with the wash buffer (20 mM Tris-HCl, pH 7.5, 10% glycerol, 300 mM NaCl, 0.5% NP-40) for 3 times, 10 min each time. Lastly, 10 μl 5 × SDS-PAGE buffer was added into the beads, and boiled at 100°C for 10 min, followed by western blot.

Phos-tag acrylamide (Vazyme, PA101) gel analysis was carried out to detect the phosphorylation of macroH2A. The Mono-nucleosome IP samples eluted by boiling were run on 8% polyacrylamide gel containing 25 μM Phos-tag acrylamide and 25 μM $MnCl_2$. After electrophoresis, Phos-tag acrylamide gels were washed with gentle shaking in transfer buffer containing 1 mM EDTA for 15 min and then incubated in transfer buffer containing 0.01% SDS without EDTA for 15 min. Proteins were then transferred to PVDF membrane, which was incubated with anti-macroH2A primary antibodies, followed by standard western blot protocol. The uncropped and unprocessed scans of all of the blots shown in the Source Data file.

Western blot analysis

The cells were harvested in lysis buffer (50 mM Tris-HCl, pH 7.4, 150 mM NaCl, 1 mM EDTA, 1% Triton X-100) and boiled for 10 min at 100°C. Whole cell extracts (20 μg protein) were separated by SDS-PAGE and electrotransferred onto a PVDF membrane (Millipore, ISEQ00010). After blocking with 5% nonfat milk for 1 hour, the membrane was incubated with primary antibodies overnight at 4°C and washed 3 times for 10 min each with TBST, after which the membrane was incubated with the corresponding HRP-conjugated secondary antibodies for 1 hour at room temperature. The reaction was visualized using Pierce Western ECL substrate (Thermo Fisher Scientific, 32109) and detected by exposure to gel documentation system. The antibodies used included: anti-macroH2A1.2 antibodies (Cell Signaling Technology, 4827, 1:1000), anti-SPT16 antibodies (Cell Signaling Technology, 12191, 1:1000), anti-SSRP1 antibodies (Biolegend, 609702, 1:1000), anti-HA antibodies (Abcam, ab9110, 1:1000), anti-H2A antibodies (Abcam, ab18255, 1:1000), anti-H3 antibodies (ABclonal, A2348, 1:3000), anti-actin antibodies (ABclonal, AC026, 1:20000), goat anti-rabbit second antibody (Zhongshan Golden Bridge Bio-technology, ZB2301, 1:50000), and goat anti-mouse second antibody (Zhongshan Golden Bridge Bio-technology, ZB-2305, 1:50000).

Cell migration assays

Briefly, an 8 μm pore-size invasion chamber (Corning, 3422) was used for the migration assay. 3 × 10^4 cells were washed once in serum-free medium and inoculated into the upper chamber, and the lower compartment was filled with of complete medium 20% FBS. After 48 hours, the migration cells adhering to the bottom surface of the chamber membrane were stained with 0.2% crystal violet. Images of three different fields were captured from each membrane, and the number of migration cells was counted using the ImageJ image processing program.

Cell phagocytosis assay

For the assessment of cell phagocytosis by flow cytometry, the cells were seeded in 6-well plates at 5×10^5 cells/well, and then PE-labeled beads (Thermo Fisher Scientific, F13083) was added to each well at the working concentration. After an incubation period of 24 hours, the medium was removed and the cells were washed thoroughly 3 times. 15,000 cells from each group were analyzed via flow cytometry and phagocytic active (PE^+) macrophages was determined. The proportion of PE^+ cell number to the total cell number was calculated.

The cell phagocytosis was also assessed by using phagocytosis video shooting in IncuCyte.[49] Briefly, the cells were seeded in 24-well plates at 10^4 cells/well, cultured, and treated as described above. The cell plate was transferred into IncuCyte S3 (Sartorius) immediately after PE-labeled beads was added to each well at the working concentration. A time series module was chosen, and 360 cycles of the same field was shot for 15 min per cycle with a × 20 objective. Phagocytosis video shooting was performed with IncuCyte S3 integrated software (Incucyte 2022A).

QUANTIFICATION AND STATISTICAL ANALYSIS

All statistical analyses were performed in GraphPad Prism (version 8.0.1). The outcomes of all statistical tests including *p* values and number of samples are included in the figure panels or the corresponding figure legends. Significance was defined as any statistical outcome that resulted in a *p* value of less than 0.05.